

Support-Change-Tolerant Continuity for Averaged Categorical Shared-Exclusions PID

Closed-simplex total-variation bounds, exact counterexamples,
and a finite-sample transfer without a positive support-mass floor

pid-rs project analysis

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Abstract

Categorical shared-exclusions partial information decomposition (SxPID) is defined pointwise and then averaged with the same joint law. Existing interior differentiability and fixed-support arguments do not by themselves control this average when cells enter or leave the support. This document gives a project-defined exact-real continuity theorem on a fixed finite ambient alphabet. A pointwise overlap decomposition separates residual entropy from a common-overlap neighborhood load. For an event that is a union of J equivalence neighborhoods, the load is at most J times total variation and yields the modulus

$$(1 - \eta) \log \left(1 + \frac{J\eta}{1 - \eta} \right).$$

The construction is sharp for this decomposition and is realizable by a paper-semantic Sx event. Published nonnegativity of the informative and misinformative pointwise atoms sharpens the full-lattice Möbius transfer: component residuals use a maximum of two residual entropies, whereas signed net residuals require their sum. Finite-alphabet entropy envelopes produce an explicit modulus that vanishes without a positive population support-mass floor. Monotone relaxations compose with a dependency-colored empirical-law radius. Exact categorical counterexamples reject a global linear modulus, pointwise boundary continuity, an active-face Fannes–Audenaert substitution, a maximum-residual net shortcut, and alphabet-free control. Lean checks the exact heterogeneous keyed Sx events and the finite equivalence-union load bound, but not the complete logarithmic and averaged theorem. The result does not prove binary64 enclosure, estimator calibration, or continuous PID validity.

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1 Status, scope, and provenance

The categorical Sx functional, redundancy lattice, and pointwise component-sign theorem are due to Makkeh, Gutknecht, and Wibral [1, 2]. This document analyzes that paper-defined functional. The support-change theorem and its validation artifacts are project-defined. They do not define another PID measure and make no scientific-priority claim.

Fix nonempty finite source alphabets, a nonempty finite target alphabet, the complete Cartesian-product alphabet, and the full finite redundancy lattice. These objects and the event map remain fixed while the law changes. All logarithms are natural and all mathematical statements use exact real arithmetic.

$$\mathcal{Z} = \mathcal{S}_1 \times \cdots \times \mathcal{S}_m \times \mathcal{T}, \quad K = |\mathcal{Z}|. \quad (1)$$

The result applies to averaged categorical cumulatives and atoms. It does not apply to:

- a changing observed alphabet or selected lattice;
- pointwise values at a key whose mass vanishes;
- a same-sample fitted or changing quantizer;
- continuous, mixed, or singular shared exclusions;
- portable binary64 signs or correctly rounded logarithms; or
- statistical or consumer validity without a separate contract.

2 Overlap and residuals

Let p, q be probability laws on one finite set X . Put

$$\eta = d_{\text{TV}}(p, q) = \frac{1}{2} \|p - q\|_1, \quad (2)$$

and define

$$r_x = \min\{p_x, q_x\}, \quad a_x = p_x - r_x, \quad b_x = q_x - r_x. \quad (3)$$

Then

$$p = r + a, \quad q = r + b, \quad (4)$$

$$\sum_x a_x = \sum_x b_x = \eta, \quad \sum_x r_x = 1 - \eta =: R, \quad (5)$$

and $a_x b_x = 0$ for every x . For any law functional F , write

$$\Delta F = F(p) - F(q). \quad (6)$$

For a nonnegative finite vector d , define its residual entropy with the continuous zero convention:

$$E(d) = - \sum_{x:d_x>0} d_x \log d_x. \quad (7)$$

Write

$$E_\vee = \max\{E(a), E(b)\}, \quad E_\Sigma = E(a) + E(b). \quad (8)$$

These are entropies of subprobability vectors, not entropies of separately normalized laws.

3 Anchored-neighborhood continuity

Fix a set $N_x \subseteq X$ for every x , subject to the anchor condition

$$x \in N_x. \quad (9)$$

For a nonnegative vector v , write

$$v(N_x) = \sum_{y \in N_x} v_y. \quad (10)$$

Define

$$G_N(p) = - \sum_{x:p_x>0} p_x \log p(N_x). \quad (11)$$

Equation (9) makes every displayed logarithm well-defined.

For $R > 0$, define the overlap load

$$T_N(r, d) = \sum_{x:r_x>0} r_x \frac{d(N_x)}{r(N_x)}. \quad (12)$$

Theorem 3.1 (Residual-plus-load bound). *For any two probability laws on X ,*

$$|G_N(p) - G_N(q)| \leq E_\vee + R \log \left(1 + \frac{\max\{T_N(r, a), T_N(r, b)\}}{R} \right), \quad (13)$$

where the logarithmic term is defined as zero when $R = 0$.

Proof. Substitution gives the exact identity

$$G_N(p) - G_N(q) = C_N + U_{p,a} - U_{q,b}, \quad (14)$$

where

$$C_N = \sum_{x:r_x>0} r_x \log \frac{r(N_x) + b(N_x)}{r(N_x) + a(N_x)}, \quad (15)$$

and

$$U_{p,a} = - \sum_{x:a_x>0} a_x \log p(N_x), \quad U_{q,b} = - \sum_{x:b_x>0} b_x \log q(N_x). \quad (16)$$

The anchor condition gives

$$a_x \leq p(N_x) \leq 1, \quad b_x \leq q(N_x) \leq 1. \quad (17)$$

Consequently,

$$0 \leq U_{p,a} \leq E(a), \quad 0 \leq U_{q,b} \leq E(b), \quad (18)$$

and

$$|U_{p,a} - U_{q,b}| \leq E_\vee. \quad (19)$$

For $d \in \{a, b\}$, let

$$C_N(d) = \sum_{x:r_x>0} r_x \log\left(1 + \frac{d(N_x)}{r(N_x)}\right). \quad (20)$$

Then $C_N = C_N(b) - C_N(a)$. Both terms are nonnegative, so

$$|C_N| \leq \max\{C_N(a), C_N(b)\}. \quad (21)$$

Jensen's inequality with weights r_x/R gives

$$C_N(d) \leq R \log\left(1 + \frac{T_N(r, d)}{R}\right). \quad (22)$$

Combining the bounds proves Equation (13). If $R = 0$, the common sum is empty. If $\eta = 0$, both residuals vanish. Thus the stated convention covers both endpoints. $\square$

4 Union-of-equivalence load

Fix an integer $J \geq 1$ and suppose

$$N_x = \bigcup_{j=1}^J [x]_j, \quad (23)$$

where, for each j , $[x]_j$ is the equivalence class of x under a fixed relation $\sim_j$. The J relations may differ.

Lemma 4.1 (Fractional-cover load). *If d is nonnegative and $\sum_x d_x = \eta$, then*

$$T_N(r, d) \leq J\eta. \quad (24)$$

Proof. For $r_x > 0$,

$$\frac{d(N_x)}{r(N_x)} \leq \sum_{j=1}^J \frac{d([x]_j)}{r([x]_j)}. \quad (25)$$

Fix j and group x by an equivalence class D . Every class with $r(D) > 0$ contributes

$$\sum_{x \in D} r_x \frac{d(D)}{r(D)} = d(D). \quad (26)$$

The positive-overlap classes contribute at most η . A class with no overlap has no anchored x in the defining sum and need not contribute its residual mass. Summing the J relations proves the result. $\square$

Define

$$g_J(\eta) = \begin{cases} (1 - \eta) \log\left(1 + \frac{J\eta}{1 - \eta}\right), & 0 < \eta < 1, \\ 0, & \eta \in \{0, 1\}. \end{cases} \quad (27)$$

Corollary 4.2 (Equivalence-union bound). *Under Equation (23),*

$$|G_N(p) - G_N(q)| \leq E_V + g_J(\eta). \quad (28)$$

Moreover,

$$g_J(\eta) \leq J\eta. \quad (29)$$

Proof. Apply Lemma 4.1 in Theorem 3.1. The linear envelope follows from $\log(1 + x) \leq x$. $\square$

Remark 4.3 (Upper radii). The exact function g_J is not monotone on all of $[0, 1]$. A proof that knows only $\eta \leq \varepsilon$ cannot substitute ε blindly into Equation (27). Equation (29) is a safe monotone relaxation.

Proposition 4.4 (Paper-semantic equality witness). *For every $J \geq 1$ and $0 \leq \eta \leq 1$, there is a categorical Sx event for which*

$$G_N(p) - G_N(q) = E(a) + g_J(\eta). \quad (30)$$

Proof. Use J sources and states

$$z, \quad c, \quad \ell_1, \dots, \ell_J. \quad (31)$$

For source j , make c and ℓ_j share one categorical value. Give every other state-coordinate pair a distinct value. The singleton antichain has

$$N_{\ell_j} = \{\ell_j, c\}, \quad N_z = \{z\}, \quad N_c = \{c, \ell_1, \dots, \ell_J\}. \quad (32)$$

Let the target be constant and set

$$r_{\ell_j} = \frac{1 - \eta}{J}, \quad a_z = \eta, \quad b_c = \eta. \quad (33)$$

The residual at z contributes $E(a) = \eta \log(1/\eta)$. The residual at c contributes zero. Each leaf contributes its share of g_J . Direct summation proves the equality. $\square$

5 Categorical shared-exclusions specialization

Specialize the generic finite set to $X = \mathcal{Z}$.

Let β be a node of the full finite redundancy lattice. For every ambient key $z = (s_1, \dots, s_m, t) \in \mathcal{Z}$, define

$$A_\beta(z) = \bigcup_{u \in \beta} \{y : y_u = s_u\}, \quad (34)$$

$$C(z) = \{y : y_T = t\}, \quad B_\beta(z) = A_\beta(z) \cap C(z). \quad (35)$$

These are the finite-alphabet event semantics in Equations (15a)–(15b) and Appendix B of Makkeh, Gutknecht, and Wibral [1]. The notation makes the law-independent keyed event map explicit; it does not alter the paper-defined functional. If $J_\beta = |\beta|$, then A_β and B_β are unions of J_β equivalence neighborhoods. The target event is one equivalence neighborhood. The event map is fixed on the complete alphabet, independently of the law. Local logarithms and averages below evaluate these events only at positive-mass keys.

Support-restricted averaging gives

$$I_\beta^+(p) = G_{A_\beta}(p), \quad I_\beta^-(p) = G_{B_\beta}(p) - G_C(p), \quad (36)$$

$$I_\beta^{\text{sx}}(p) = G_{A_\beta}(p) - G_{B_\beta}(p) + G_C(p). \quad (37)$$

Theorem 5.1 (Averaged cumulative bounds). *For every full-lattice node β ,*

$$|\Delta I_\beta^+| \leq E_V + g_{J_\beta}(\eta), \quad (38)$$

$$|\Delta I_\beta^-| \leq E_V + g_{J_\beta}(\eta) + g_1(\eta), \quad (39)$$

$$|\Delta I_\beta^{\text{sx}}| \leq E_\Sigma + 2g_{J_\beta}(\eta) + g_1(\eta). \quad (40)$$

Proof. For every supported key,

$$0 \leq c_\beta^+(z; p) \leq -\log p_z, \quad 0 \leq c_\beta^-(z; p) \leq -\log p_z, \quad (41)$$

and

$$|c_\beta^{\text{sx}}(z; p)| \leq -\log p_z. \quad (42)$$

$z \in B_\beta(z) \subseteq C(z)$ and $z \in A_\beta(z)$, so each keyed event probability is at least p_z , while every event probability is at most one. The informative residual averages therefore differ by at most E_V . Treating the misinformative expression in Equation (36) as one local value gives the same E_V , not two residual entropies. Its common term has one B_β and one C contribution. The signed net residual averages require $E(a) + E(b)$; its common term has A_β , B_β , and C contributions. Apply Corollary 4.2 to each contribution. $\square$

6 Full-lattice atom transfer

Let M be the fixed Möbius inverse in the orientation

$$\Pi_\alpha^u = \sum_\beta M_{\alpha\beta} I_\beta^u. \quad (43)$$

Define

$$W_\alpha(\eta) = \sum_\beta |M_{\alpha\beta}| g_{J_\beta}(\eta), \quad (44)$$

and

$$s_\alpha = \sum_\beta M_{\alpha\beta}. \quad (45)$$

Let $\perp$ be the unique least node and Z the down-set zeta matrix. Since

$$Ze_\perp = \mathbf{1}, \quad (46)$$

we have

$$M\mathbf{1} = e_\perp, \quad s_\alpha = \mathbf{1}\{\alpha = \perp\}. \quad (47)$$

Equation (19), Möbius inversion, and Theorem IV.3 of the defining paper [1] give

$$\pi_\alpha^+(z; p) \geq 0, \quad \pi_\alpha^-(z; p) \geq 0. \quad (48)$$

They reconstruct the greatest-node component cumulative. At that node, for a supported key $z = (s, t)$,

$$c_\top^+(z; p) = -\log p(S = s) \leq -\log p_z, \quad c_\top^-(z; p) = -\log p(S = s \mid T = t) \leq -\log p_z. \quad (49)$$

Consequently, for every atom,

$$0 \leq \pi_\alpha^+(z; p) \leq -\log p_z, \quad 0 \leq \pi_\alpha^-(z; p) \leq -\log p_z. \quad (50)$$

Theorem 6.1 (Averaged atom bounds). *For every atom α of the full finite redundancy lattice,*

$$|\Delta \Pi_\alpha^+| \leq E_V + W_\alpha(\eta), \quad (51)$$

$$|\Delta \Pi_\alpha^-| \leq E_V + W_\alpha(\eta) + |s_\alpha| g_1(\eta), \quad (52)$$

$$|\Delta \Pi_\alpha^{\text{sx}}| \leq E_\Sigma + 2W_\alpha(\eta) + |s_\alpha| g_1(\eta). \quad (53)$$

Proof. Split each averaged atom into overlap, left-residual, and right-residual weighted point-wise atoms. For the overlap part, commute the fixed finite Möbius matrix with the finite sum. The A_β and B_β common terms receive the row coefficients $M_{\alpha\beta}$. Equation (47) gives the coefficient of the common target term.

For a component atom, Equations (48) and (50) place each residual average in $[0, E(a)]$ or $[0, E(b)]$. Their difference is at most $E_\vee$. For a net atom,

$$\pi_\alpha^{\text{sx}} = \pi_\alpha^+ - \pi_\alpha^-, \quad (54)$$

so

$$|\pi_\alpha^{\text{sx}}(z; p)| \leq -\log p_z. \quad (55)$$

The two signed residual averages require E_Σ . The common-term bounds give the remaining terms in Equations (51)–(53). $\square$

Remark 6.2 (Why the full lattice matters). Equation (47) requires a least node. The sign theorem in Equation (48) also concerns the paper-defined full lattice. Neither fact transfers automatically to a truncated node family or an arbitrary coefficient matrix.

7 Finite-alphabet envelopes and uniform continuity

Assume $\eta > 0$, and put

$$k_a = |\text{supp } a|, \quad k_b = |\text{supp } b|. \quad (56)$$

The residual supports are disjoint, so $k_a + k_b \leq K$. Concavity of $-x \log x$ gives

$$E(a) \leq \eta \log \frac{k_a}{\eta}, \quad E(b) \leq \eta \log \frac{k_b}{\eta}. \quad (57)$$

Therefore

$$E_\vee \leq e_K^\vee(\eta) := \eta \log \frac{K-1}{\eta}, \quad (58)$$

and

$$E_\Sigma \leq e_K^\Sigma(\eta) := \eta \log \frac{\lfloor K^2/4 \rfloor}{\eta^2}. \quad (59)$$

Define both envelopes as zero at $\eta = 0$. If $K = 1$, then $\eta = 0$.

Corollary 7.1 (Closed-simplex continuity). *On a fixed finite Cartesian-product alphabet and full finite redundancy lattice, every averaged categorical shared-exclusions cumulative and atom is uniformly continuous in total variation on the closed probability simplex. The result remains valid when cells enter or leave the support and requires no positive minimum cell mass.*

Proof. Substitute Equations (58) and (59) into Theorems 5.1 and 6.1. Every resulting term tends to zero as $\eta \downarrow 0$. The finite family of nodes can be bounded simultaneously by taking its maximum. $\square$

7.1 Range caps

Let

$$K_S = \prod_{i=1}^m |\mathcal{S}_i|. \quad (60)$$

The local ranges give

$$0 \leq I_\beta^+(p) \leq H_p(S) \leq \log K_S, \quad (61)$$

$$0 \leq I_\beta^-(p) \leq H_p(S | T) \leq \log K_S. \quad (62)$$

The same upper bounds hold for every averaged component atom because nonnegative component atoms sum to the greatest-node component cumulative. Hence every component difference can be capped by $\log K_S$, and every net difference can be capped by $2 \log K_S$.

8 Support-change-tolerant statistical transfer

The exact functions $g_J, e_K^\vee, e_K^\Sigma$ are not all monotone on $[0, 1]$. To use only an upper radius $0 \leq \eta \leq \varepsilon \leq 1$, define

$$\bar{e}_K^\vee(\varepsilon) = \varepsilon \left(1 + \log \frac{K-1}{\varepsilon} \right), \quad (63)$$

$$\bar{e}_K^\Sigma(\varepsilon) = \varepsilon \left(2 + \log \frac{\lfloor K^2/4 \rfloor}{\varepsilon^2} \right), \quad (64)$$

with value zero at $\varepsilon = 0$. These are monotone on $[0, 1]$, and

$$E_V \leq \bar{e}_K^\vee(\varepsilon), \quad E_\Sigma \leq \bar{e}_K^\Sigma(\varepsilon), \quad g_J(\eta) \leq J\varepsilon. \quad (65)$$

To verify the first two inequalities, put $A = K-1 \geq 1$ and $B = \lfloor K^2/4 \rfloor \geq 1$. For $0 < \eta \leq \varepsilon$, both $\log(A/\varepsilon)$ and $\log(B/\varepsilon^2)$ are nonnegative, while

$$\eta \log \frac{\varepsilon}{\eta} = \varepsilon \left(\frac{\eta}{\varepsilon} \right) \log \left(\frac{\varepsilon}{\eta} \right) \leq \frac{\varepsilon}{e} \leq \varepsilon. \quad (66)$$

Therefore

$$\eta \log \frac{A}{\eta} \leq \varepsilon \log \frac{A}{\varepsilon} + \varepsilon, \quad (67)$$

and

$$\eta \log \frac{B}{\eta^2} \leq \varepsilon \log \frac{B}{\varepsilon^2} + 2\varepsilon. \quad (68)$$

The endpoint $\eta = 0$ follows by the zero convention. The last inequality follows from $g_J(\eta) \leq J\eta \leq J\varepsilon$.

Put

$$L_\alpha = \sum_\beta |M_{\alpha\beta}| J_\beta. \quad (69)$$

On any event where $d_{TV}(\hat{p}, p) \leq \varepsilon$, Table 1 holds.

Table 1: Monotone support-change transfer from a total-variation radius.

Quantity	Radius
I_β^+	$\bar{e}_K^\vee(\varepsilon) + J_\beta \varepsilon$
I_β^-	$\bar{e}_K^\vee(\varepsilon) + (J_\beta + 1)\varepsilon$
I_β^{sx}	$\bar{e}_K^\Sigma(\varepsilon) + (2J_\beta + 1)\varepsilon$
Π_α^+	$\bar{e}_K^\vee(\varepsilon) + L_\alpha \varepsilon$
Π_α^-	$\bar{e}_K^\vee(\varepsilon) + (L_\alpha + s_\alpha)\varepsilon$
Π_α^{sx}	$\bar{e}_K^\Sigma(\varepsilon) + (2L_\alpha + s_\alpha)\varepsilon$

The dependency-colored empirical-law theorem in the companion pid-rs analysis supplies an L^1 radius D_n under a declared mutual-within-color independence contract. Set

$$\varepsilon_n = D_n/2. \quad (70)$$

Table 1 then gives an averaged categorical SxPID radius without a population support-mass floor. The composition does not validate the coloring from data and does not remove the exponential alphabet factor in the law-distance theorem.

For a nonidentical-law theorem centered at $\bar{p}_n$, let

$$b_n = \|\bar{p}_n - p_\star\|_1. \quad (71)$$

If D_n controls $\|\hat{p}_n - \bar{p}_n\|_1$, use

$$\varepsilon_n = \min \left\{ 1, \frac{D_n + b_n}{2} \right\}. \quad (72)$$

An unknown b_n remains an unknown scientific bias.

9 Exact negative results

Counterexample 9.1 (No global linear or pointwise boundary modulus). Let S_2 be constant and $T = S_1$. Give the rare copied state probability η and compare with the point mass at $\eta = 0$. The averaged unique- S_1 informative and net atom is

$$h_2(\eta) = -\eta \log \eta - (1 - \eta) \log(1 - \eta). \quad (73)$$

The two-source bottom event is the whole sample space because the S_2 branch is constant, so its local cumulative is zero. Möbius inversion therefore makes the unique- S_1 atom equal to the S_1 -node cumulative. The node's net local value is pointwise mutual information, which reduces to source surprisal when $T = S_1$; its averaged cumulative is $I(S_1; T)$. The L^1 distance is 2η , and

$$\frac{h_2(\eta)}{2\eta} \longrightarrow \infty. \quad (74)$$

The rare-key pointwise atom is $-\log \eta$. Therefore no global linear modulus exists, the $\eta \log(1/\eta)$ order is necessary, and pointwise support-boundary continuity is false.

Counterexample 9.2 (An active-face Fannes–Audenaert substitution fails). Use three sources, a constant target, the singleton antichain, and source states

$$000, \quad 011, \quad 202, \quad 330, \quad 444. \quad (75)$$

Set

$$p = (0, 3/10, 3/10, 3/10, 1/10), \quad (76)$$

$$q = (1/10, 3/10, 3/10, 3/10, 0). \quad (77)$$

Then $\eta = 1/10$ and

$$|G_N(p) - G_N(q)| = \frac{1}{10} \log 10 + \frac{9}{10} \log \frac{4}{3}. \quad (78)$$

The union-support face has

$$K_{\cup} = |\text{supp } p \cup \text{supp } q| = 5. \quad (79)$$

The ordinary Fannes–Audenaert entropy radius on that five-state face is

$$h_2(1/10) + \frac{1}{10} \log 4. \quad (80)$$

It is smaller. The exact excess is

$$\frac{9}{10} \log \frac{6}{5} - \frac{1}{10} \log 4 > 0, \quad (81)$$

equivalently

$$6^9 > 4 \cdot 5^9. \quad (82)$$

This is an actual categorical Sx event. It rejects substituting the ordinary entropy continuity radius computed on the union-support face for the neighborhood-functional bound, and it attains the residual-plus- g_3 bound. It does not say that the same expression formed with a larger ambient Cartesian-product cardinality must fail. A bounded search found no four-state example; that non-finding does not prove global minimality.

Counterexample 9.3 (The signed residual maximum shortcut fails). Keep S_2 constant. For $0 < \eta < 1$ and $R = 1 - \eta$, put

$$p(0, 0) = \eta, \quad p(1, 2) = p(2, 1) = R/2, \quad (83)$$

$$q(1, 1) = \eta, \quad q(1, 2) = q(2, 1) = R/2. \quad (84)$$

The displayed pairs are (S_1, T) . As above, the constant- S_2 branch makes the bottom net cumulative zero, so the two-source Möbius inversion identifies the unique- S_1 net atom with the S_1 -node cumulative. The node's local value is pointwise mutual information, while its averaged cumulative is $I(S_1; T)$. The difference of the two residual-weighted contributions is

$$2\eta \log \frac{1+\eta}{2\eta} > \eta \log \frac{1}{\eta}. \quad (85)$$

The strict inequality reduces to $(1 - \eta)^2 > 0$. Thus the signed residual budgets cannot be replaced by their maximum in this proof. The example does not prove global sharpness of the whole net-atom modulus.

Counterexample 9.4 (A truncated-family row identity fails). For a V-shaped three-element poset with two incomparable minimal nodes and one greatest node, constant cumulatives $c = (1, 1, 1)$ invert to

$$\pi = (1, 1, -1). \quad (86)$$

The greatest-row Möbius sum is -1 . The full-lattice row-sum and component-sign arguments cannot be transferred to an arbitrary truncated family.

Counterexample 9.5 (No alphabet-free modulus). Move total mass η from one cell and spread it uniformly over M new cells in a copied source construction. The averaged unique component contains $\eta \log M$. A modulus independent of the fixed ambient alphabet cannot hold.

Proposition 9.6 (Worst-case leading logarithmic coefficients). *There exist fixed finite categorical shared-exclusions systems and fixed atom coordinates whose component and net differences have leading coefficients one and two, respectively, in units of $\eta \log(1/\eta)$ as $\eta \downarrow 0$. Consequently, consider a family that assigns every covered fixed system $\mathcal{F}$ a bound $c\eta \log(1/\eta) + O_{\mathcal{F}}(\eta)$, where the lower-order constant may depend on $\mathcal{F}$ but c is common to the family. Such a family requires $c \geq 1$ for components and $c \geq 2$ for net atoms.*

Proof. For fixed $K \geq 2$, the upper entropy terms satisfy

$$e_K^{\vee}(\eta) = \eta \log \frac{1}{\eta} + O(\eta), \quad e_K^{\Sigma}(\eta) = 2\eta \log \frac{1}{\eta} + O(\eta), \quad (87)$$

and $g_J(\eta) = O(\eta)$ for fixed J . Hence every W_α and row-sum-weighted g_1 term is $O(\eta)$ when the finite lattice, Möbius matrix, and branch counts are fixed.

The copied-source path gives

$$\lim_{\eta \downarrow 0} \frac{h_2(\eta)}{\eta \log(1/\eta)} = 1. \quad (88)$$

This gives one fixed system and atom with informative-component coefficient one. The constant-target equality witness has identical informative and misinformative bottom atoms and gives one fixed-system misinformative witness with the same coefficient.

For the fixed two-source system in the signed-residual counterexample, the complete unique- S_1 net atom is $I(S_1; T)$. With $R = 1 - \eta$, direct evaluation gives

$$I_p(S_1; T) = -\eta \log \eta - R \log \frac{R}{2}, \quad (89)$$

and

$$I_q(S_1; T) = \eta \log \frac{4\eta}{(1+\eta)^2} + R \log \frac{2}{1+\eta}. \quad (90)$$

Subtracting gives

$$\begin{aligned} |\Pi_{U_1}^{\text{sx}}(p) - \Pi_{U_1}^{\text{sx}}(q)| &= 2\eta \log \frac{1+\eta}{2\eta} \\ &\quad + (1-\eta) \log \frac{1+\eta}{1-\eta}. \end{aligned} \quad (91)$$

Both terms are positive for $0 < \eta < 1$. Dividing by $\eta \log(1/\eta)$ and taking $\eta \downarrow 0$ gives

$$\lim_{\eta \downarrow 0} \frac{|\Pi_{U_1}^{\text{sx}}(p) - \Pi_{U_1}^{\text{sx}}(q)|}{\eta \log(1/\eta)} = 2. \quad (92)$$

This gives one fixed system and atom with net coefficient two. $\square$

Remark 9.7 (Sharpness scope). Proposition 9.6 establishes only the common worst-case leading coefficients for families of fixed-system bounds whose $O_{\mathcal{F}}(\eta)$ remainder may depend on the system. It does not claim that every system or atom attains either coefficient, and it does not make the lower-order terms, branch terms, or every individual cumulative and atom bound globally sharp.

10 Formal and executable evidence boundary

The Lean artifact checks finite-vector overlap and residual algebra, residual-entropy ceilings, abstract component and signed transfer lemmas, and a generic down-set Möbius row-sum identity. It also defines the exact heterogeneous dependent Cartesian key, the keyed shared-exclusions events A_β , B_β , and C , their equivalence-class covers, anchors, intersection identity, positive event masses, and law-independent event map. A separate checked module proves the support-restricted finite load bound $0 \leq T_N \leq J\eta$, including its concrete A_β , B_β , and C corollaries. Lean also checks endpoint and linear properties of g_J .

Lean does not yet bundle probability laws or semantic total variation, define the support-restricted logarithmic functional, derive g_J from T_N , identify the complete redundancy lattice and its concrete Möbius inverse, import the published pointwise sign theorem, or compose these facts into averaged Sx cumulatives and atoms. Sampling, Rust refinement, and binary64 arithmetic also remain outside the checked theorem.

The implementation-separated generator is standard-library-only and imports no pid-rs code. It retains rational laws, exact integer identities, high-precision logarithmic references, equality witnesses, and a bounded seeded challenge corpus. The larger historical grid searches remain separate revision-1 route evidence. This separation reduces shared implementation risk; it is not independent authorship or review. Decimal comparison is not certified interval arithmetic. The Rust comparison replays committed empirical count tables. It is not a universal refinement proof or a portable binary64 enclosure.

The dependency-colored probability theorem has a separate premise: complete rows in each declared color must be mutually independent. The code cannot infer that premise from a finite sample.

11 Release interpretation

The exact-real mathematical result supports this statement:

Averaged categorical shared-exclusions cumulatives and full-lattice atoms are quantitatively continuous on a fixed finite closed simplex, including across support changes, without a positive minimum cell-mass premise.

It does not support these statements:

- every pointwise atom has a finite support-boundary limit;
- the functional is globally Lipschitz;
- the result is independent of alphabet size;
- a binary64 atom sign is certified;
- a dependence coloring is valid because autocorrelation is small;
- continuous PID2 is generally consistent;
- full continuous PID3 is ready for scientific inference; or
- a PID value can independently authorize an action.

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